

Practical No: 02

Aim: To demonstrate various pre-data processing techniques for a given dataset.

Software Requirement: Python IDE, Google Colab, VS Code

Theory:

Introduction

Data preprocessing is the initial and one of the most critical steps in any data science or machine learning workflow. Raw data gathered from various sources may contain missing values, inconsistent formats, duplicates, or irrelevant fields. Without proper preprocessing, even the most advanced models may produce inaccurate results.

The goal of preprocessing is to transform raw data into a structured and cleaned format, ready for analysis and modeling.

Creation and Loading of Dataset

Before applying preprocessing techniques, the data must first be created or loaded. In Python, datasets are typically loaded using the pandas library, which provides the DataFrame structure.

Common formats include:

- Manual creation using Python dictionaries or lists
- External data files like .csv, .xlsx, .json, etc.

1. Reshaping the Data

Reshaping means altering the structure of the data to suit specific analytical needs. Sometimes data may be in a “wide” format but needs to be converted to “long” format, or vice versa, for better analysis.

Reshaping techniques help reformat and organize data logically.

Functions Mentioned:

- `pd.DataFrame()` – to create a dataset manually

- `df` – A variable that stores the DataFrame created from the data dictionary.
- `df` inside `print(df)` – Displays the full content of the DataFrame in tabular form.

These transformations make it easier to analyze time-series or hierarchical data.

2. Filtering the Data

Filtering involves retrieving specific rows or columns from a dataset that meet certain conditions. This is useful for removing irrelevant data and focusing on particular subsets.

Filtering also helps in identifying outliers or values that meet thresholds.

Functions Mentioned:

`df.filter(['Name'])` – Filters the DataFrame to only include the 'Name' column.

3. Merging Datasets

Often, information is split across multiple datasets. Merging helps in combining these datasets based on a common identifier or key.

Types of merges include:

- Inner Join
- Left Join
- Right Join
- Outer Join

Functions Mentioned:

- `pd.merge(df1, df2, on='key')` – merges two datasets
- `df.join()` – merges based on index

Concept:

- `details` contains student info
- `fees_status` contains fee details

- `merge()` combines both based on the ID column

Merging is essential in building a complete dataset for analysis or training.

4. Handling Missing Values

Missing values can occur due to human error, equipment malfunction, or data corruption. These need to be addressed before analysis.

Techniques:

- **Removal:** Drop rows or columns with missing values.
- **Imputation:** Replace with:
 - Mean / Median / Mode
 - Constant value
 - Forward-fill / Backward-fill

Functions Mentioned:

- `np.nan`– Represents a missing or null value in numeric computations.
- `df[G2].mean()`– Calculates the mean of column G2, ignoring NaN values.
- `df[G2].fillna(value)`– Fills missing values (NaN) with a specified value (e.g., mean).
- `df.loc[:, 'G2'] = ...` – Updates the entire column G2 with the new filled data.
- `print(df)`– Displays the final, updated DataFrame.

Correct handling ensures the model is trained on reliable data.

5. Normalization / Feature Scaling

Normalization is used to bring features into the same scale or range. This is important because many algorithms are sensitive to the magnitude of data.

Functions Mentioned:

- `MinMaxScaler()` – Initializes a scaler that maps data into [0, 1] range.
- `scaler.fit_transform(df)`– Fits the scaler to the data and transforms it in one step.

Scaling improves model accuracy, especially for distance-based algorithms like KNN.

Importance of Data Preprocessing

Without Preprocessing

Irregular, messy data

Biased or failed results

Model overfitting/underfitting

Poor training speed

With Preprocessing

Clean, structured data

Reliable insights

Improved model performance

Fast and efficient modeling

Conclusion

Hence I have successfully demonstrated various pre-data processing techniques for a given dataset.